

PYTHON LOGICAL QUESTIONS — LOOPS

Q1

```
for i in range(5):  
    print(i)
```

Q2

```
for i in range(1, 6):  
    print(i)
```

Q3

```
for i in range(2, 10, 2):  
    print(i)
```

Q4

```
i = 0  
while i < 3:  
    print(i)  
    i += 1
```

Q5

```
for i in range(5, 0, -1):  
    print(i)
```

Q6

```
for i in range(1, 10):  
    if i == 5:  
        break
```

```
print(i)
```

Q7

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)
```

Q8

```
i = 5  
while i > 0:  
    print(i)  
    i -= 2
```

Q9

```
for i in range(3):  
    for j in range(2):  
        print(i, j)
```

Q10

```
for i in range(1, 4):  
    for j in range(i):  
        print("*", end="")  
    print()
```

Q11

```
for i in range(3):  
    for j in range(3):
```

```
    if i == j:
        print("X", end="")
    else:
        print("O", end="")
print()
```

Q12

```
for i in range(1, 6):
    if i % 2 == 0:
        continue
    print(i)
```

Q13

```
i = 1
while i <= 10:
    if i == 6:
        break
    print(i)
    i += 2
```

Q14

```
for i in range(3):
    print(i)
else:
    print("Done")
```

Q15

```
for i in range(3):
```

```
    if i == 1:
        break
    print(i)
else:
    print("Done")
```

Q16

```
i = 0
while i < 3:
    print(i)
```

👉 Will this terminate? Why or why not?

Q17

```
i = 1
while i <= 5:
    print(i)
    i = i + i
```

Q18

```
i = 10
while i:
    print(i)
    i //= 3
```

Q19

```
for i in range(1, 5):
    for j in range(1, i+1):
        print(j, end="")
```

```
print()
```

Q20

```
for i in range(5):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

```
else:
```

```
    print("End")
```

Q21

```
i = 0
```

```
while i < 5:
```

```
    if i == 2:
```

```
        i += 1
```

```
        continue
```

```
    print(i)
```

```
    i += 1
```

Q22

```
for i in range(1, 4):
```

```
    for j in range(1, 4):
```

```
        if i + j == 4:
```

```
            break
```

```
        print(i, j)
```

Q23

```
for i in range(3):
```

```
for j in range(3):  
    if j == 1:  
        continue  
    print(i, j)
```

Q24

```
i = 1  
while i < 20:  
    print(i)  
    i *= 2
```

Q25

```
for i in range(10, 0, -3):  
    print(i)
```

PRACTICE QUESTIONS — LOOPS (PYTHON)

Q1

Write a program to print numbers from **1 to 10** using a for loop.

Q2

Write a program to print numbers from **10 to 1** using a while loop.

Q3

Write a program to print **even numbers from 1 to 20**.

Q4

Write a program to print the **multiplication table of a given number**.

Q5

Write a program to calculate the **sum of numbers from 1 to n**.

Q6

Write a program to count the **number of digits** in a number.

Q7

Write a program to **reverse a number**.

Q8

Write a program to check whether a number is a **palindrome**.

Q9

Write a program to find the **factorial of a number**.

Q10

Write a program to print the **Fibonacci series** up to n terms.

Q11

Write a program to print all numbers from 1 to 50 that are **divisible by 3**.

Q12

Write a program to count how many numbers between 1 and 100 are **even and odd**.

Q13

Write a program to check whether a number is **prime**.

Q14

Write a program to find the **sum of digits** of a number.

Q15

Write a program to find the **largest digit** in a number.

Q16

Write a program to print this pattern:

```
*  
  
**  
  
***  
  
****
```

Q17

Write a program to print this pattern:

```
1  
12  
123  
1234
```

Q18

Write a program to print this pattern:

```
1  
22  
333  
4444
```

Q19

Write a program to print this pattern:

**

*

Q20

Write a program to print a **square of stars** of size n.

Q21

Write a program to check whether a number is an **Armstrong number**.

Q22

Write a program to find the **sum of factorials of digits** of a number.

Q23

Write a program to print all **prime numbers between 1 and 100**.

Q24

Write a program to find the **GCD (HCF)** of two numbers using loops.

Q25

Write a program to find the **LCM** of two numbers using loops.

Q26

Write a program to count how many **zeros** are present in a number.

Q27

Write a program to remove **all digits except even digits** from a number.

Q28

Write a program to check whether a number is a **perfect number**.

Q29

Write a program to print **numbers whose digits are in increasing order**.

Q30

Write a program to simulate a **number guessing game** (1–10).

ADVANCED & HARD LOOP PRACTICE — PYTHON

Q 1 Count Special Numbers

Print numbers from **1 to N** that satisfy **ALL**:

- Divisible by 3
 - Not divisible by 5
 - Sum of digits is even
-

Q 2 Reverse Pattern Pyramid

Print:

12345

1234

123

12

1

Q 3 Longest Digit Sequence

Given a number, find the **longest sequence of increasing consecutive digits**.

Q 4 Digital Root Using Loop

Find the **digital root** of a number using loops only
(repeat sum of digits until single digit).

Q 5 Count Prime Digits

Count how many digits in a number are **prime digits** (2,3,5,7).

Q 6

Predict output:

```
for i in range(1, 6):
```

```
    if i % 2 == 0:
```

```
        break
```

```
    print(i)
```

Q 7

Predict output:

```
i = 1
```

```
while i <= 10:
```

```
    if i == 5:
```

```
        i += 2
```

```
        continue
```

```
    print(i)
```

```
    i += 3
```

Q 8

Predict output:

```
for i in range(3):
```

```
for j in range(3):  
    if i == j:  
        continue  
    print(i, j)
```

Q 9

Predict output:

```
i = 1  
while i < 20:  
    if i % 3 == 0:  
        i += 1  
        continue  
    print(i)  
    i *= 2
```

Q 10 (VERY TRICKY)

Predict output:

```
for i in range(1, 5):  
    for j in range(1, 5):  
        if i * j > 6:  
            break  
        print(i, j)
```

Q 1 1 Print Number Diamond

```
1  
121  
12321  
121
```

1

Q 1 2 Frequency Counter

Count frequency of **each digit** in a number.

Q 1 3 Mirror Number

Check whether a number becomes **the same after reversing twice** (*no string conversion*).

Q 1 4 Pattern Logic Trap

Predict output:

```
for i in range(5):
    for j in range(i):
        print(i, end=" ")
    print()
```

Q 1 5 Half Pyramid with Logic

Print:

```
*
* *
* * *
* * * *
```

Q 1 6 Strong Number

Check if a number is **Strong** (sum of factorial of digits = number).

Q 1 7 Twin Primes

Print all **twin primes** between 1 and 100.

Q 1 8 Rotational Equality

Check whether a number remains the same after **rotating digits k times**.

Q 1 9 Loop-Only Sort

Sort a list of numbers **using loops only** (no built-in sort).

Q 2 0 Hidden Pattern

Predict output:

```
i = 0
```

```
while i < 5:
```

```
    j = 0
```

```
    while j < i:
```

```
        print(i, end="")
```

```
        j += 1
```

```
    i += 1
```

```
    print()
```